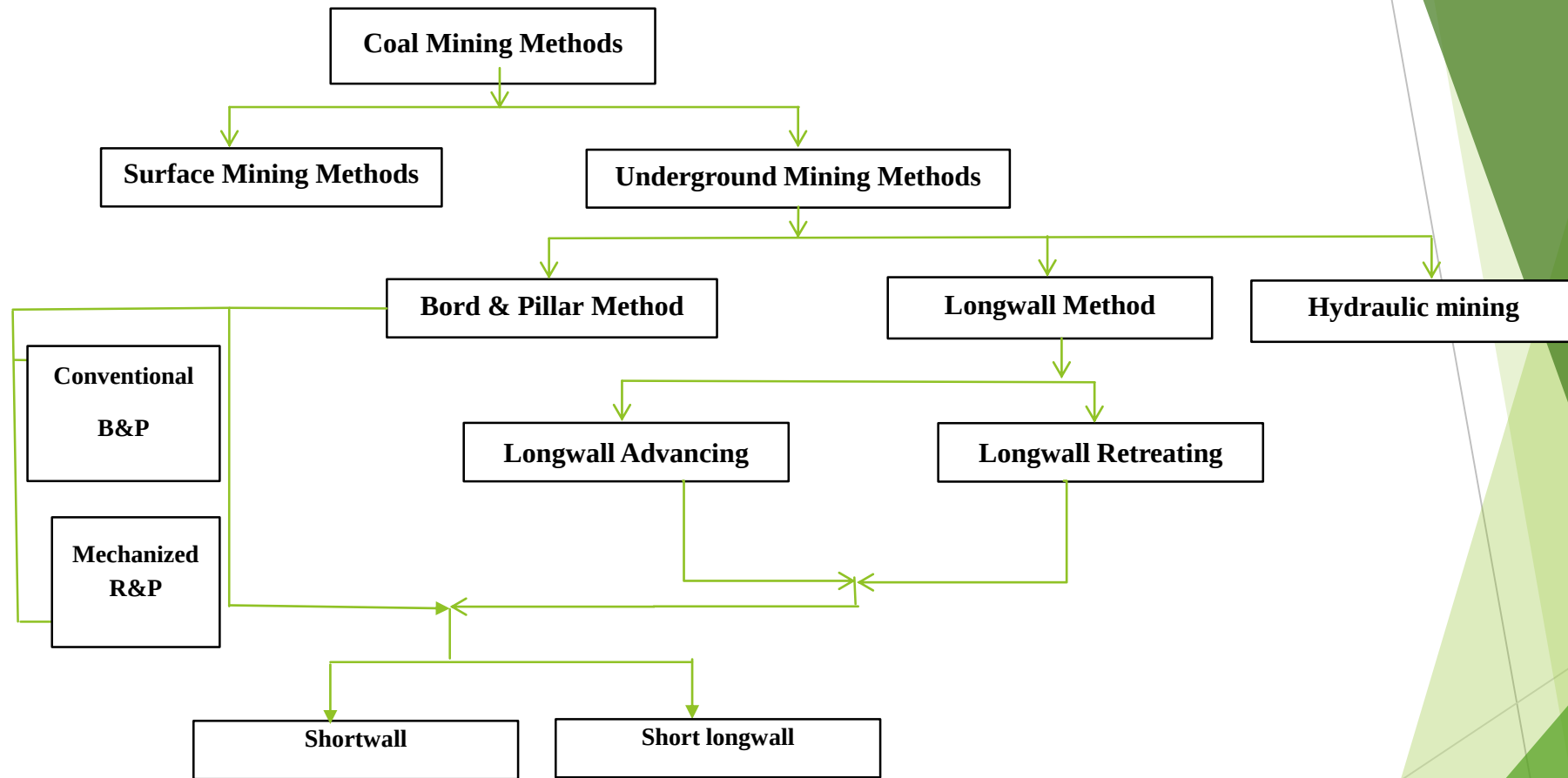


Underground Mining Methods

R M Bhattacharjee

Mining Methods

- Opencast
- Underground
 - Inseam mining
 - Horizon mining
 - Bord & Pillar / Room & Pillar
 - Longwall mining



Basic methods of mining

Factors affecting choice of mining methods

▶ Physical

- ▶ Thickness of seam
- ▶ Gradient of seam
- ▶ Depth of seam
- ▶ Shape of deposit
- ▶ Structure and geological anomalies
- ▶ Gassiness
- ▶ Proneness to spontaneous heating
- ▶ Characteristics of roof and floor rocks
- ▶ Hydrogeological conditions

Factors affecting choice of mining methods

▶ Technical

- ▶ Technological development
- ▶ Availability of machines
- ▶ Whether in abundance or short supply
- ▶ Total reserves available in the mine
- ▶ Infrastructure available

▶ Commercial

- ▶ Capital fund
- ▶ Running capital
- ▶ Coal price
- ▶ Cost on import

Factors affecting choice of mining methods

► Socio-economic

- Demand for coal
- Employment opportunities
- Skill available
- Availability of land and other resources
- Policy for import of coal

Comparison of methods

Parameters	B&P	R&P	LW
% extraction	Very poor	High	Very high
Continuous / Cyclic mining	Cyclic	Mostly continuous	Continuous
Fleeting of face machinery	Very high	Requires fleeting intermittently	No fleeting of face machinery
Subsidence control	Not simple and uniform	Mostly uniform	Simple and uniform
Strata control	Not very effective with conventional supports / roof bolts	Largely effective with high capacity bolts, cable bolts and goaf edge supports	Efficient strata control with powered supports
Ventilation	Sluggish due to numerous connections & leakage points and scattered workings	Better ventilation due to concentrated workings, use of auxiliary ventilators	Efficient ventilation due to longer pillar, less connections and concentrated workings

Comparison of methods

Parameters	B&P	R&P	LW
Risk of spontaneous heating	Very high due to loss of coal, lack of goaf packing, slow rate of workings leading to delay in panel sealing	Relatively less risk due to high rate of production, better extraction percentage, better goaf packing	Less risk of fire due to insignificant loss of coal, good caving and packing, scope of inertization
Productivity	Poor	Better	Very high
Mass production	Not possible	Substantial improvement in volume of production	Mass production possible
Scope of mechanization	Little	Medium	Very high
Concentration of activities - miners and machines	Scattered	Mostly concentrated	Concentrated

Inseam mining Vs Horizon mining

Inseam mining

- ▶ The seam is accessed through shaft or incline or adit
- ▶ Development of coal seam is done following the seam
- ▶ Once the seam is touched by the access entries, further development of all roads, haulage & ventilation, are driven along the seam
- ▶ Most of the mines are developed by following inseam mining

Horizon mining

- ▶ Seam is accessed from the shaft or incline through level roads, cross cuts in stone
- ▶ The seam is accessed at various levels through drivage in stones
- ▶ The deposit between the levels are exploited subsequently



Inseam mining

Applicability:

- ▶ Flat seams or low dip seams
- ▶ Relatively Free from faults
- ▶ Medium to high thickness
- ▶ Seam density is low
- ▶ Seam occurring at shallow to medium depth
- ▶ Size of mine is small



Inseam mining

Advantage:

- ▶ Does not require development through stone, which is costly and time taking
- ▶ Gestation period is low
- ▶ Seams can be developed in any order, ordinarily in descending order

Inseam mining

Disadvantages

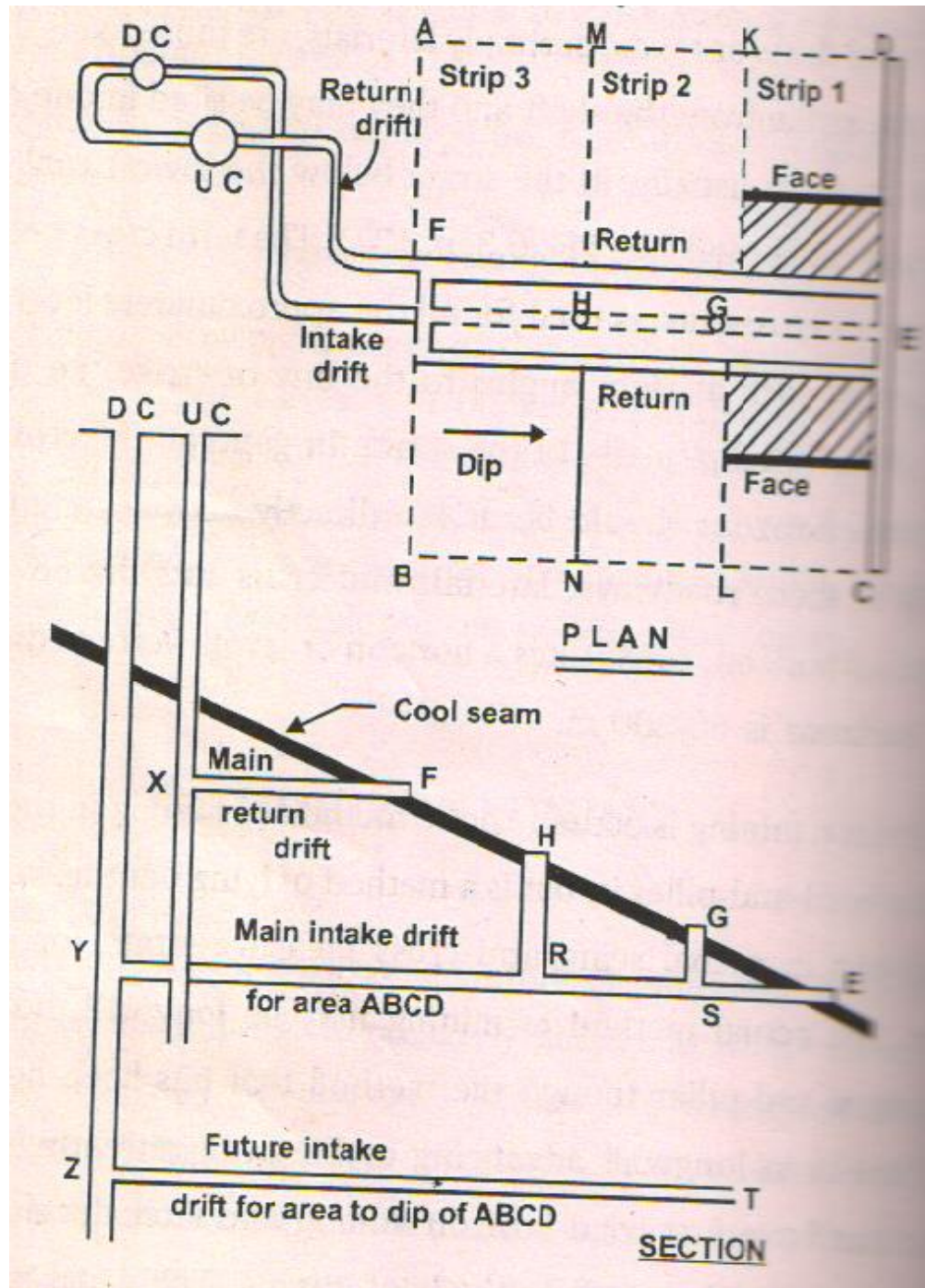
- ▶ For irregular deposit, having number of faults, inclined seams, development by inseam mining is costly
- ▶ Separate drivages for haulage, ventilation etc. for each seam is costly
- ▶ Stability of such roads in coal is less compared to drivages in stone
- ▶ Capital as well as maintenance cost of pipe line, electrical cable, stowing ranges is multiplicative for each seam workings
- ▶ Large volume of production from single mine is difficult

Horizon mining layout

- ▶ Coal seams are opened up by level roads driven in rock instead of driving through seam
- ▶ Suitable for multiple seams at close proximity
- ▶ Seams with steep gradient
- ▶ Thick seams

Layout

- ▶ Main roads are level and driven through rock, not in coal
- ▶ Main levels handle coal from different seams
- ▶ Main ventilation is also through main levels
- ▶ Cross cuts from levels are driven to intersect the seams
- ▶ One level intersect different seams at different points
- ▶ These main roads are maintained throughout the life of the mine
- ▶ As driven in rock, stability is better
- ▶ Levels are opened at different horizons



Advantages

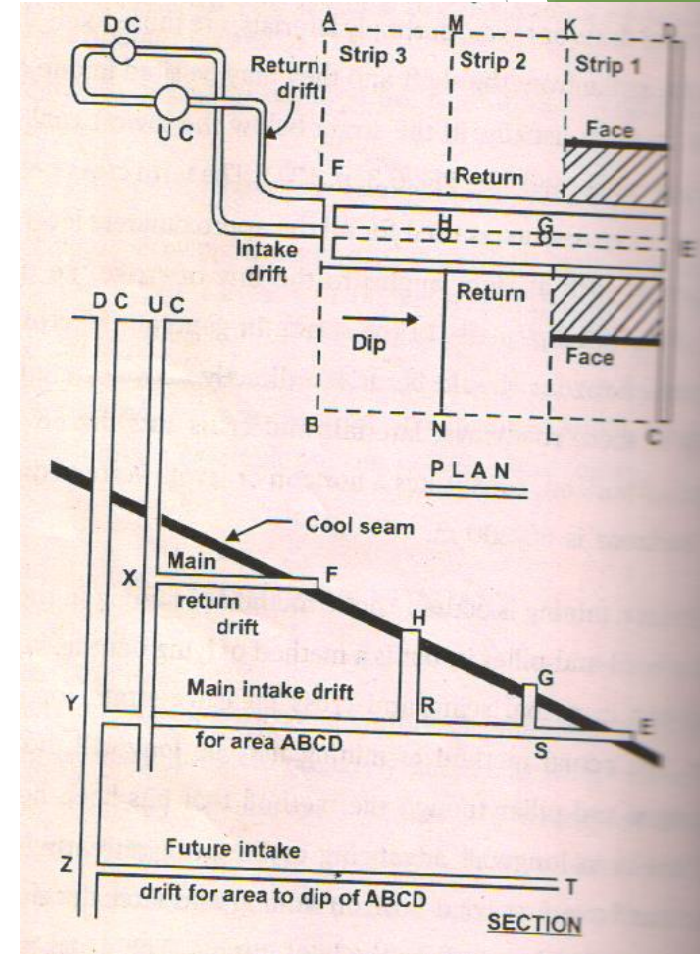
- ▶ Multiple seams may be worked out together
- ▶ Same system of transportation, ventilation, pipe lines used for all seams
- ▶ Same horizon remain in operation for long period
- ▶ Concentration of operation
- ▶ Better supervision
- ▶ High volume of production
- ▶ Stable main roads
- ▶ Use of locomotives, belt conveyors, coal haulers
- ▶ Less maintenance
- ▶ Once reserve between two horizons for all seams exhausted, the horizon may be abandoned
- ▶ No loss of coal in main roads

Disadvantages

- ▶ Longer gestation periods
- ▶ Increased cost for drivage through stone
- ▶ Horizons require to be maintained for very long time
- ▶ Not suitable for smaller reserves

Operations

- ▶ Top seam is worked first
- ▶ Remaining seams afterwards in descending order
- ▶ Main intake and main returns are set out level from respective shafts to intercept the seam at E and F
- ▶ First area to be worked ABCD. Vertical interval may vary from 100 - 150 m
- ▶ Roadways are driven in the seam to form return airways and to open out faces, conveyor roadways like ED, EC, GZ, HZ
- ▶ Two faces are opened up on two sides
- ▶ When area ABCD is worked out, another main intake is driven at a lower level to intercept seam at J in order to work out the next area.
- ▶ Intake airway for ABCD now acts as return of next area.



Bord & Pillar Mining

Dr R.M. Bhattacharjee

B & P Mining

- ▶ Driving series of roadways in the seam parallel to each other and connected by cross headings so as to form pillars
- ▶ These pillars formed are extracted later on partially or fully, as permitted by geological and other conditions
- ▶ Pillars may be square, rectangular or rhombus shaped
- ▶ Worked in two phases
 - ▶ Development - Formation of pillars
 - ▶ Depillaring - Extraction of pillars formed

B & P Mining

- ▶ Applicability:
 - ▶ Flat seams
 - ▶ Average thickness (3 to 4.8m)
 - ▶ Shallow depth

B & P Mining

▶ Advantage:

- ▶ Flexibility
- ▶ Area is proved during development
- ▶ Initial planning may be modified
- ▶ Roadways are supported by solid coal pillars
- ▶ No rigid cycle of operation every shift
- ▶ Scope of more employment
- ▶ Less capital investment

B & P Mining

- ▶ Disadvantage:
 - ▶ % extraction poor
 - ▶ Constant fleeting of machinery
 - ▶ Subsidence control is not simple
 - ▶ Strata control is difficult with conventional support system
 - ▶ Poor ventilation
 - ▶ Risk of spontaneous heating
 - ▶ Poor productivity
 - ▶ Not suitable for mass production
 - ▶ Limited scope of mechanization

B & P Mining

▶ Applicable regulation:

▶ Reg 111 of CMR 2017

- ▶ Dimension of pillars and galleries to ensure stability during the period between development and extraction
- ▶ Height not more than 3m
- ▶ Width not more than 4.8m
- ▶ Pillars normally be square or rectangular in shape
- ▶ Centre to centre distance between adjacent pillars shall not be less than that specified corresponding to the depth of workings from surface and width of galleries

B & P Mining

Depth from surface	Centre to Centre distance between adjacent pillars (in metre) for gallery width of			
	3.0m	3.6m	4.2m	4.8m
Not exceeding 60m	12.0	15.0	18.0	19.5
Exceeding 60m not exceeding 90m	13.5	16.5	19.5	21.0
Exceeding 90m not exceeding 150m	16.5	19.5	22.5	25.0
Exceeding 150m not exceeding 240m	22.5	25.5	30.5	34.5
Exceeding 240m not exceeding 360m	28.5	34.5	39.5	45.0
Exceeding 360m	39.0	42.0	45.0	48.0

Stages of B & P Mining system

- ▶ Development - Formation of pillars
- ▶ Depillaring - Extraction of pillars formed

Variants of B & P Mining system

- ▶ Develop entire area into pillars and then extract the pillars from boundary
- ▶ Develop the area into panels and extract pillars panel-wise
- ▶ Whole followed by Broken

Develop entire area into pillars and then extract

► Advantage:

- More number of working faces
- Deployment of more miners
- Quick built up of large output
- Flexibility of deployment of man and machine
- No loss of coal in barrier pillar

► Disadvantage

- Pillars standing for long time, liable to spalling, risk of premature collapse
- Sluggish ventilation
- Coal dust treatment costly and tedious
- Risk of fire or explosion spreading over entire mine
- Scattered working

Develop entire area into panels and extract panel wise

- ▶ Advantage:
 - ▶ Limited risk of loss of coal due to spontaneous heating
 - ▶ In the event of fire, the panel can be isolated from the rest of the workings
 - ▶ Risk of explosion can be limited to the panel of occurrence.
 - ▶ Crushing of pillars avoided
 - ▶ Development and depillaring can be done at the same time in different panels
 - ▶ Control of subsidence possible. Magnitude of subsidence can be reduced by working panels of sub-critical width

Develop entire area into panels and extract panel wise

► Disadvantage

- Considerable coal loss in barrier pillars
- More air-crossing for ventilation
- Separate set of equipment for each panel
- Crushing of barriers may result in joining of two adjacent panels resulting in sudden subsidence / airblast

Whole followed by Broken

- ▶ To open out the whole mine with few headings and retreat back from boundary, broken workings following whole workings
- ▶ Advantage:
 - ▶ Efficient ventilation
 - ▶ Simpler and less costly coal dust treatment
 - ▶ Intensive mechanisation possible with high OMS
 - ▶ Simpler organisation
 - ▶ Less chance of crushing and premature collapse of pillars
 - ▶ Simpler haulage system
 - ▶ Same system for development and depillaring in one panel
 - ▶ Better control of fire

Whole followed by Broken

► Disadvantage:

- Comparatively longer time for production
- Loss of coal in barrier pillars
- Separate set of equipment and manpower
- Crushing of barriers may result in joining of two adjacent panels resulting in sudden subsidence / air-blast

Design of B&P workings

- ▶ Formation of panel
 - ▶ Size of panel
 - ▶ Incubation period
 - ▶ Rate of extraction
 - ▶ Strata control consideration
 - ▶ Size of barriers
 - ▶ Strength of coal
 - ▶ Depth

Provisions of CMR, 2017

Reg. 137. - Precautions against spontaneous heating -
The following further precautions shall be taken against the danger of spontaneous heating :

- ▶ (1)(a) The seam or section shall be worked in panels having independent ventilation in such a manner that it is possible to isolate one from another easily if necessary.
- ▶ Where development has already been made without regard to this factor, artificial panels should be created by the construction of stoppings.
- ▶ In determining the size of the panel due consideration shall be given to the desirability of enabling complete extraction of the pillars therein within the incubation period of the coal.

Design of B&P workings

- ▶ Determine the size of a B&P panel with working depth of 180m from surface and producing 200 TPD.
- ▶ Assume your own conditions as necessary

Design of B&P workings

- ▶ Formation of panel
 - ▶ Size of pillars
 - ▶ Depth
 - ▶ Strength of coal
 - ▶ Nature of roof and floor
 - ▶ Geological consideration
 - ▶ Time dependent strain
 - ▶ Width of bords
 - ▶ Height of pillars

Design of B&P workings

- ▶ Mining process
 - ▶ Panel layout
 - ▶ Cyclic mining
 - ▶ Drill, blast and load
 - ▶ Cut, drill, blast and load
 - ▶ Continuous mining
 - ▶ Continuous miners, shuttle cars and conveyors

Design of B&P workings

- ▶ Mining process
 - ▶ Cutting or breaking of coal
 - ▶ Transport of coal
 - ▶ Ventilation system

Design of B&P workings

- ▶ Development
 - ▶ Manual
 - ▶ Drill and blast, manual or mechanised loading, track haulage / conveyor
 - ▶ Cut, drill & blast, manual / mechanised loading, track haulage / conveyor
 - ▶ Cut and load mechanically by CM, shuttle cars and conveyors

Design of B&P workings

- ▶ Resources to be deployed
 - ▶ Equipment
 - ▶ Manpower
 - ▶ Energy
- ▶ Safety issues
 - ▶ Stability
 - ▶ Environment
 - ▶ Operational Hazards

Face Loading

- ▶ Side Discharge Loader (SDL)
- ▶ Load Haul Dumper
- ▶ Scraper

Face Transportation

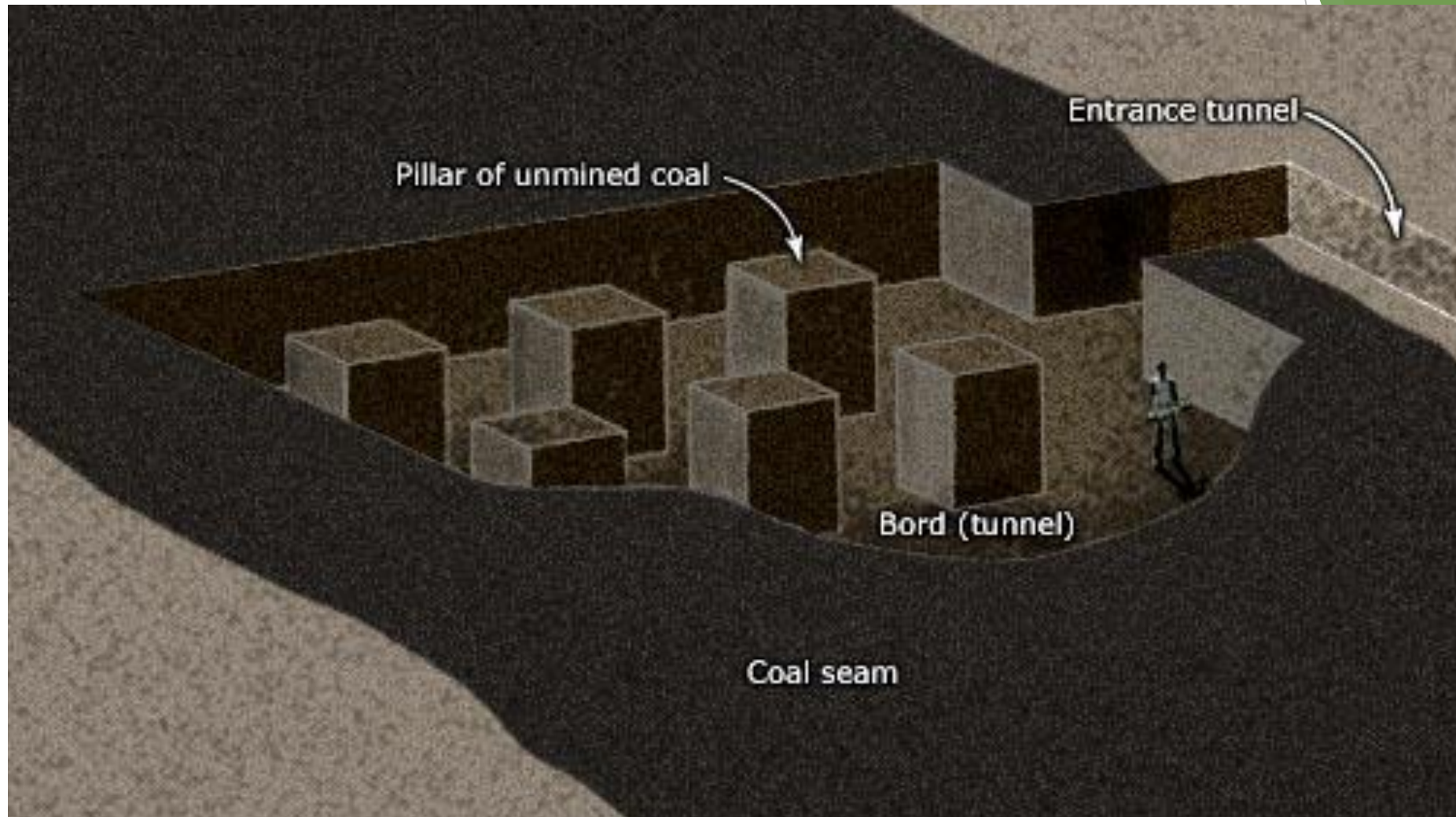
- ▶ Rope Haulage
- ▶ Chain Conveyor
- ▶ Belt conveyor

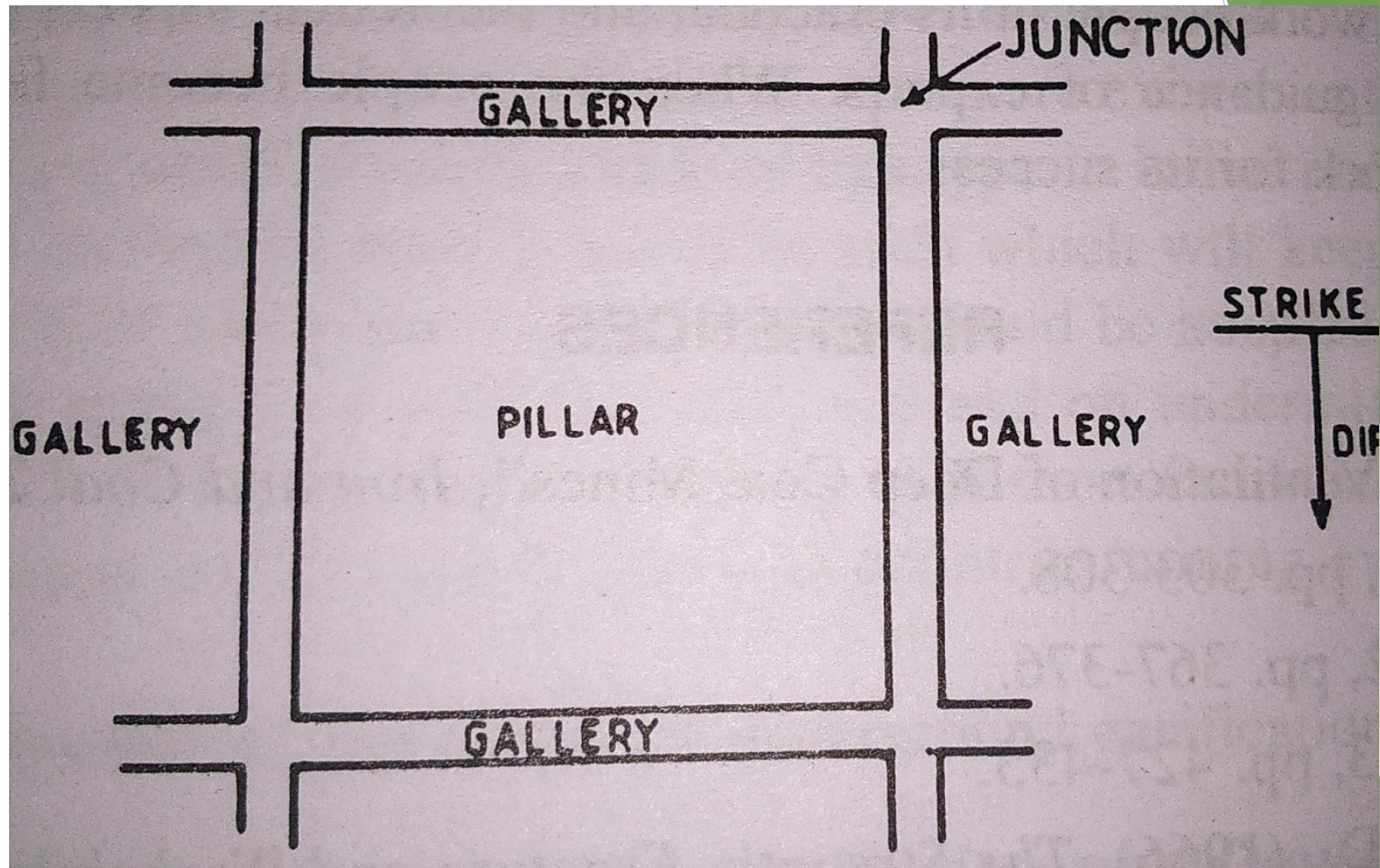
SDL



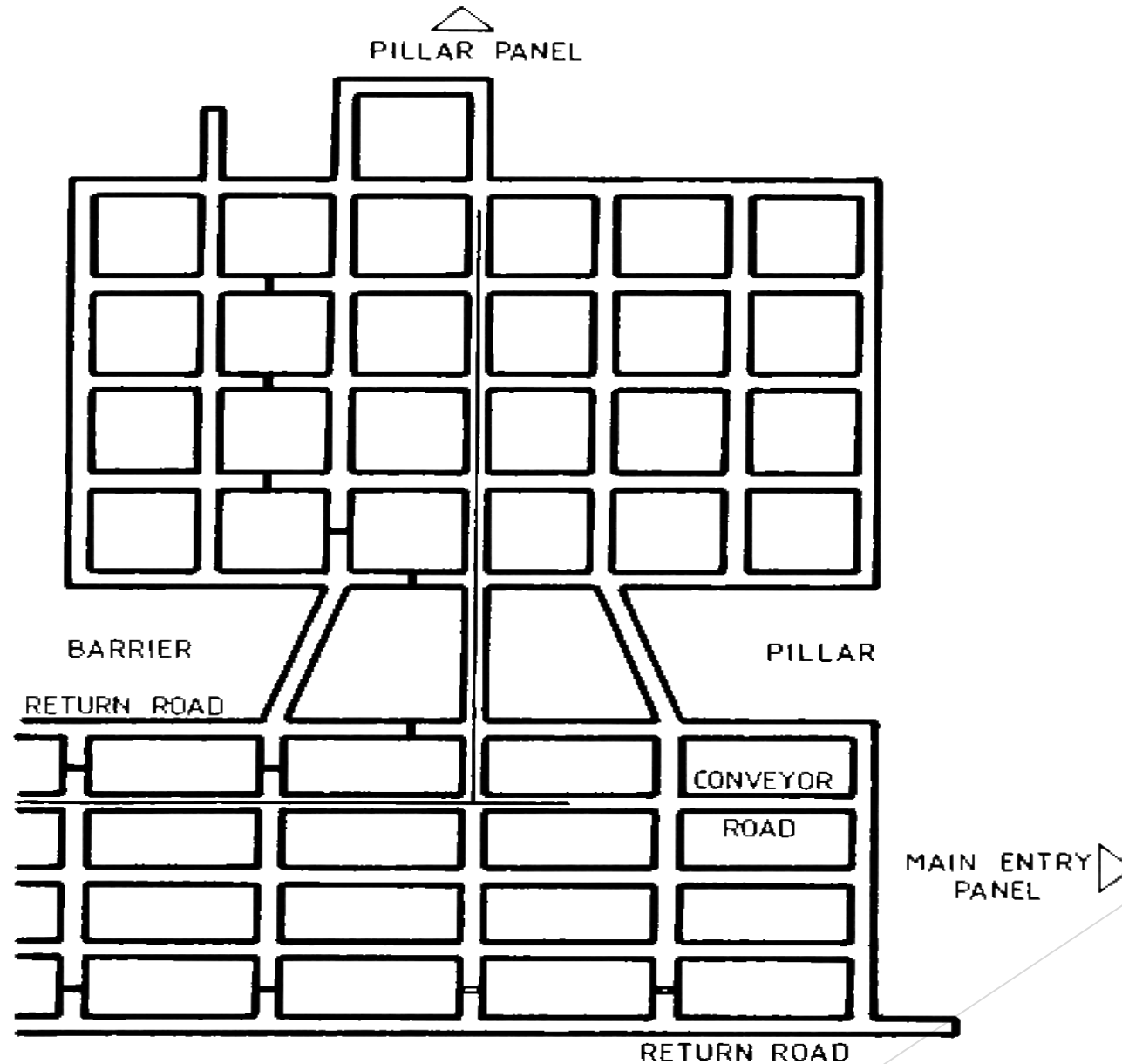
Load Haul Dumpers



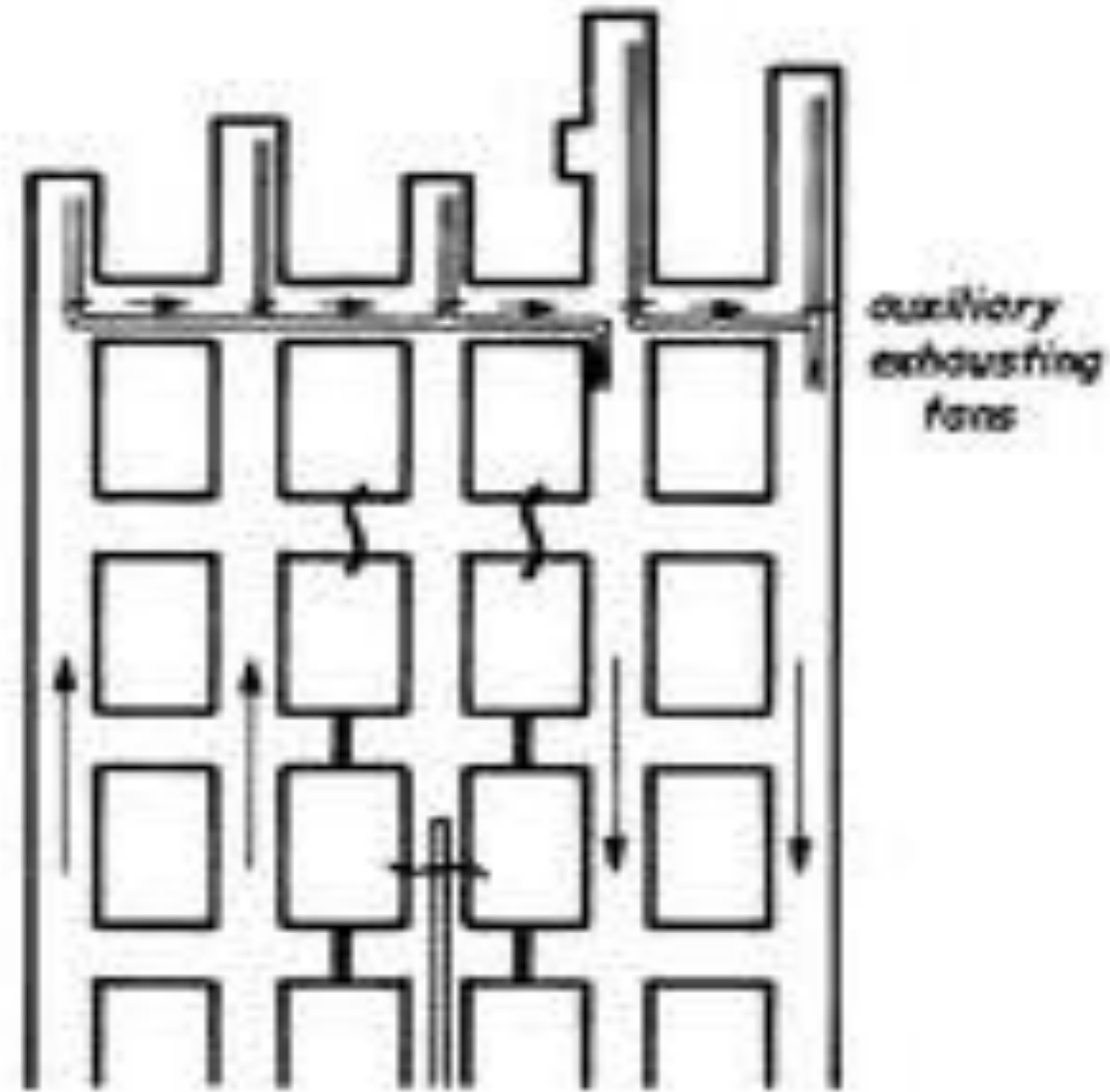


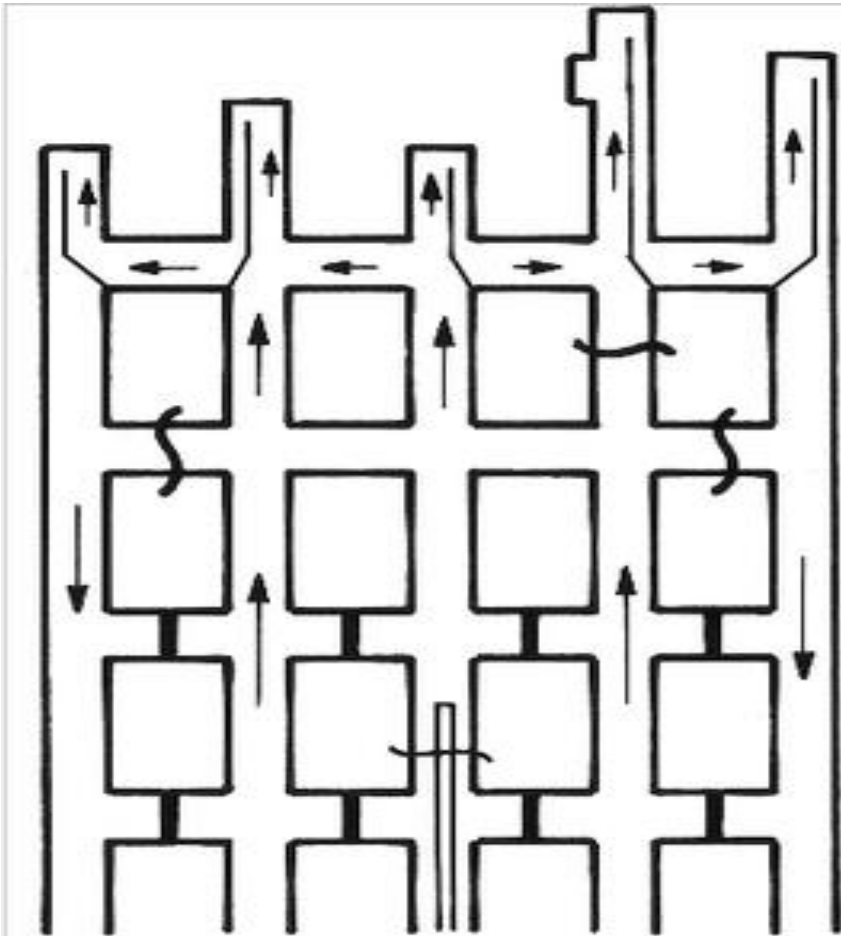


Design of B&P workings

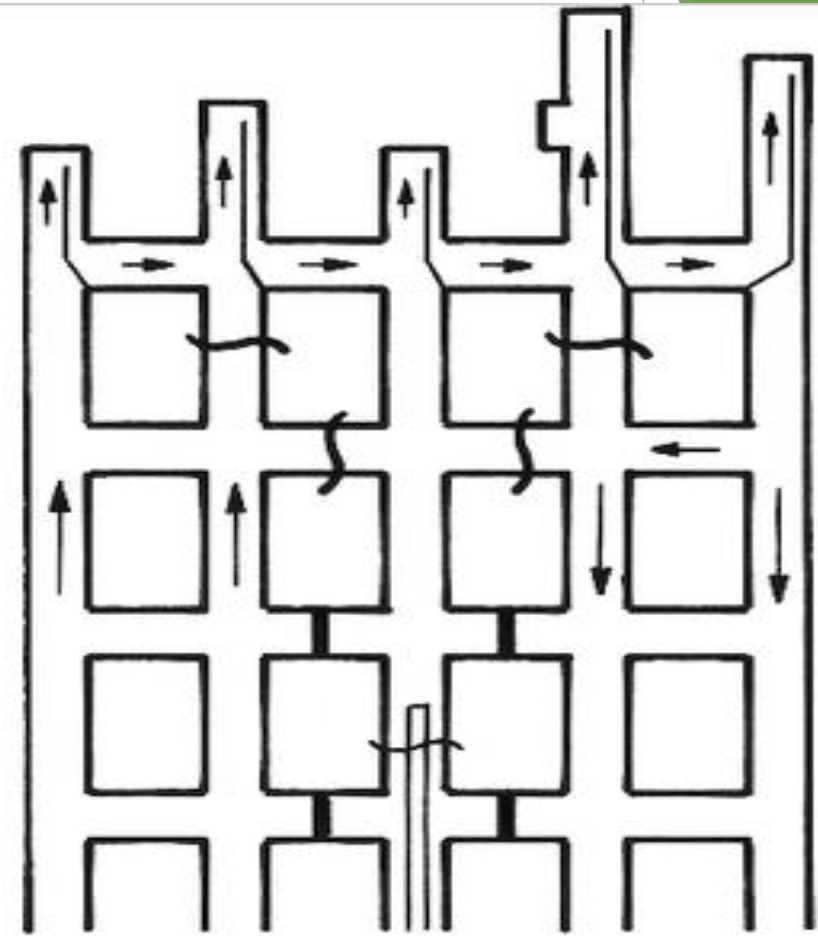


Design of B&P workings

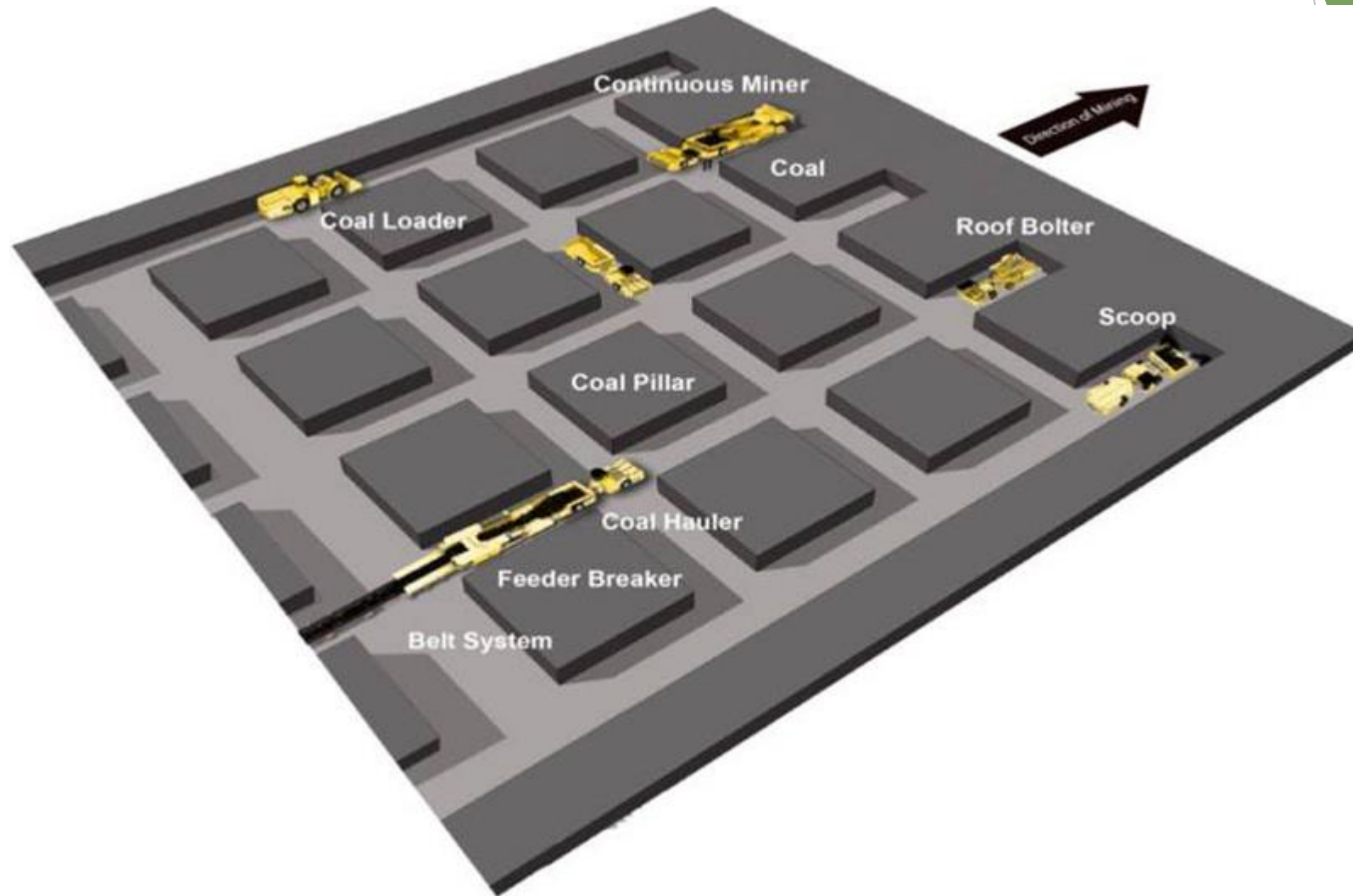




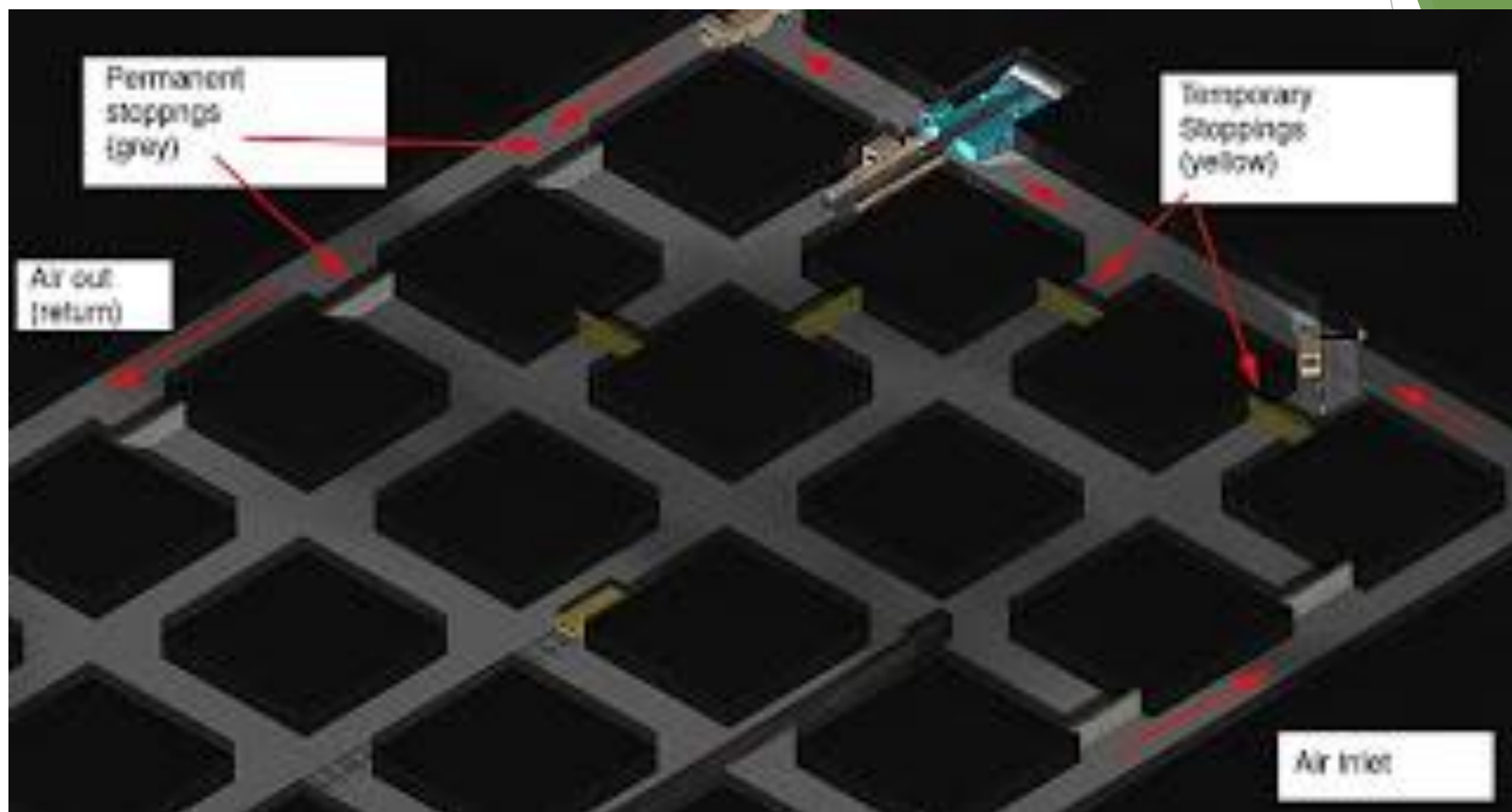
(a) Bi-directional (W) system.



(b) Uni-directional (U-tube system).



Layout of Five heading development



Percentage of extraction

Depth from surface	Centre to Centre distance between adjacent pillars (in metre) for gallery width of (%age extraction)							
	3.0m		3.6m		4.2m		4.8m	
Not exceeding 60m	12.0	43.7	15.0	42.2	18.0	41.2	19.5	43.17
Exceeding 60m not exceeding 90m	13.5	39.55	16.5	39.8	19.5	38.4	21.0	40.5
Exceeding 90m not exceeding 150m	16.5	33.06	19.5	33.5	22.5	33.8	25.0	34.0
Exceeding 150m not exceeding 240m	22.5	24.8	25.5	26.2	30.5	25.6	34.5	25.9
Exceeding 240m not exceeding 360m	28.5	19.95	34.5	19.7	39.5	20.1	45.0	20.2
Exceeding 360m	39.0	14.8	42.0	16.4	45.0	17.8	48.0	19.0



Design of B&P workings

- ▶ Extraction of pillars
 - ▶ Sequence of extraction
 - ▶ Manner of extraction
 - ▶ Sealing of a panel

Depillaring

- ▶ Extraction of pillars after formation
 - ▶ Full extraction
 - ▶ Partial extraction
 - ▶ Caving method
 - ▶ Stowing method

Problems of pillar extraction

- Strata control
 - Load on pillars
 - Overriding of pillars
 - Premature collapse
 - Bumps
 - Air blast
- Maintenance of environmental condition
 - Quantity of air
 - Gas
 - Dust
 - Illumination

Problems of pillar extraction

- ▶ Coal loss
- ▶ Spontaneous heating
- ▶ Mechanisation of coal face due to strata problem
- ▶ Poor production and productivity

Depillaring by caving method

- Coal in pillars is extracted and roof is allowed to break and collapse into the voids or the de-coaled area, known as goaf.
- As roof strata above the coal seam breaks, the ground surface develops cracks and subsides
- Extent of damage depending upon depth, thickness of the seam extracted, the nature of strata, thickness of the subsoil and effect of drag by faults.

Depillaring by caving method

Advantage

- ▶ Most popular method
- ▶ Does not require stowing
- ▶ Quicker method of exploitation
- ▶ Cheaper compared to stowing

Depillaring by caving method

Disadvantage

- Cause subsidence and may damage to surface and surface structures
- %-age of Extraction not very high
- Stability of work place critical
- Chance of spon com and heating while extraction
- Significant Fire and explosion hazard
- To have inertisation facility for successful extraction
- Requires post mining monitoring, subsidence / environment
- Very critical for geologically disturbed area

Statutory requirement for Depillaring

- Reg 100 (1)- No extraction or reduction of pillars shall be commenced, conducted or carried out except with the permission in writing of the Regional Inspector and in accordance with such conditions as he may specify therein.
- An application for permission under this sub-regulation shall be accompanied by two copies of an up-to-date plan of the area where pillars are proposed to be reduced or extracted showing the proposed extent of extraction or reduction of pillars, the manner in which such extraction or reduction is to be carried out, thickness and depth of the seam, the nature of the roof, and the rate and direction of dip.

Statutory requirement for Depillaring

- ▶ (2) The extraction or reduction of pillars shall be conducted in such a way as to prevent, as far as possible the extension of a collapse or subsidence of the goaf over pillars which have not been extracted.
- ▶ (3)(a) Save as provided by clause (b), no pillars shall be reduced or split in such a manner as to reduce the dimensions of the resultant pillars below those required by regulation 99 or by any order made there under, nor shall any gallery be so heightened as to exceed three meters.

Statutory requirement for Depillaring

- ▶ (b) During the extraction of pillars, no splitting or reduction of pillars or heightening of galleries shall be effected for a distance greater than the length of two pillars ahead of the pillar that is being extracted or reduced :

Provided that where pillar extraction is about to begin in a district such splitting or reduction of pillars or the heightening of galleries shall be restricted to a maximum of four pillars. The width of the split-galleries shall not exceed the width prescribed for galleries under regulation 99(4).

- ▶ (c) The Regional Inspector may, by an order in writing and stating the reasons therefore, relax or restrict the provisions of this sub-regulation in respect of any specified workings to such extend and on such conditions as he may specify therein.

Statutory requirement for Depillaring

- ▶ (5) Whether the method of extraction is to remove all the coal or as much of the coal as practicable and to allow the roof to cave in, the operations shall be conducted in such a way as to leave as small an area of un-collapsed roof as possible with due regard to danger from air blast or weighting on pillars. Where possible, suitable means shall be adopted to bring down the goaf at regular intervals.

Statutory requirement for Depillaring

- ▶ (6) Where the voids formed as a result of extraction are stowed with sand or other materials, the owner, agent or manager shall, on or before 10th day of every month submit to the Regional Inspector a statement giving quantity of coal raised and the quantity of sand or other materials stowed in every district during the preceding month.

Statutory requirement for Depillaring

- 118A. - Further precautions against spontaneous heating - The following further precautions shall be taken against the danger of spontaneous heating:
- (1)(a) The seam or section shall be worked in panels having independent ventilation in such a manner that it is possible to isolate one from another easily if necessary. Where development has already been made without regard to this factor, artificial panels should be created by the construction of stoppings. In determining the size of the panel due consideration shall be given to the desirability of enabling complete extraction of the pillars therein within the incubation period of the coal.

Statutory requirement for Depillaring

- ▶ 118A (1) (b) No coal, shale or other carbonaceous material shall be left or stacked belowground. Where removal of fallen coal out of the mine is not practicable, the area shall be effectively sealed off.
- ▶ (d) A panel is isolated by adequate stoppings as soon as it has been goaved out.

Statutory requirement for Depillaring

- 118A(3)(c)- Every depillaring district shall be inspected on every idle day, and all **unused working including unused working which have not been sealed off** shall be inspected once atleast in every seven days, by a competent person for any fire risk. The isolation stopping built around goaved out areas and unused workings shall be inspected at similar intervals. A report of every such inspection shall be recorded in a bound paged book kept for the purpose, and shall be signed and dated by the person making the inspection.

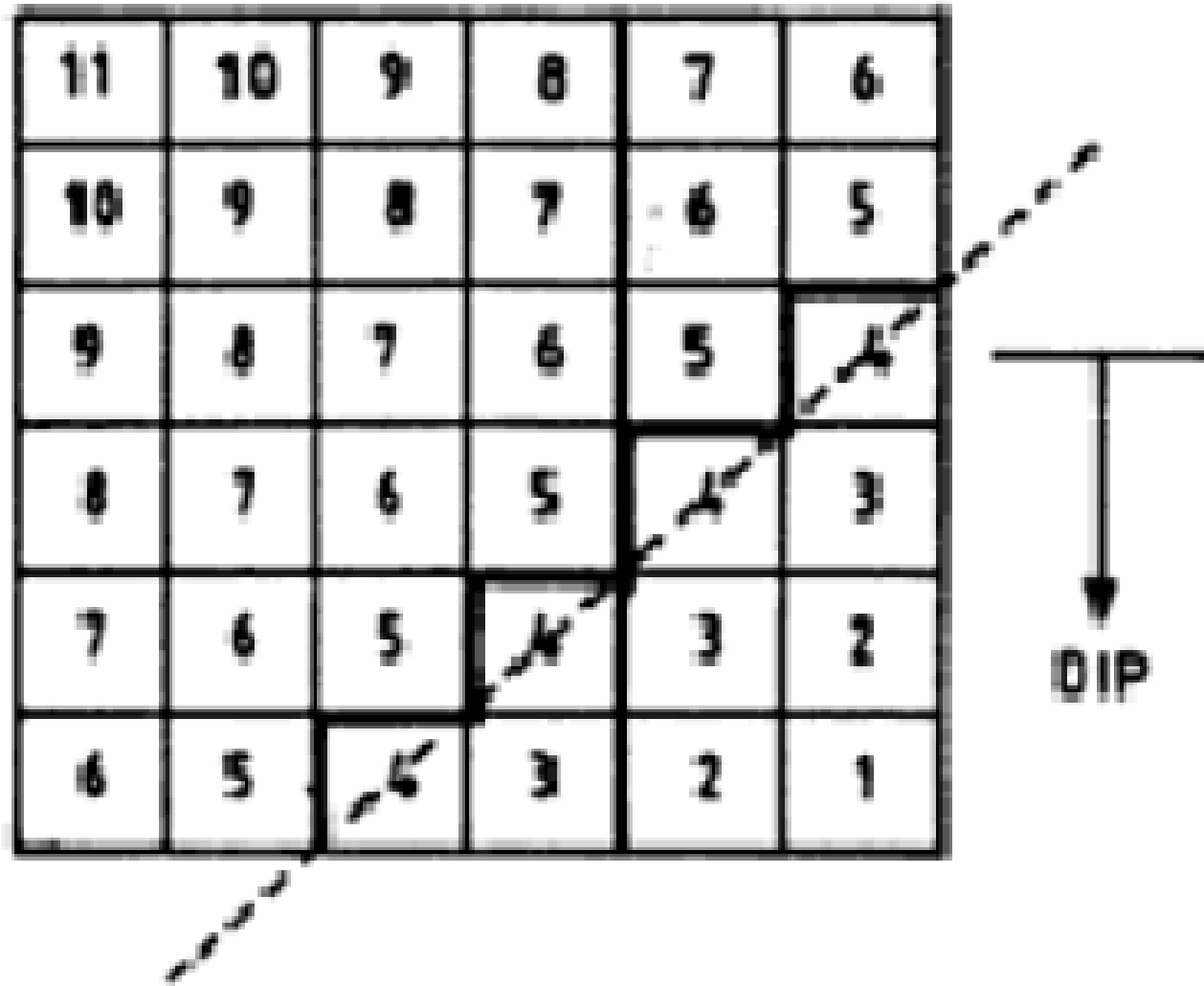
Principles of Pillar extraction techniques

- ▶ Roof exposure at one time should be minimal. In the Indian coalfields, where caving is practiced, 60-90 m² exposure is normally allowed. But in stowing districts the exposure may be increases up to 90-100 m².
- ▶ The size of the panel should be such as depillaring can be completed within the incubation period.
- ▶ Inertisation may help in extending the panel extraction time and controlling spon com hazard
- ▶ The extraction line should be so arranged as to facilitate roof control. In practice diagonal line, or step diagonal line of face is common. In special cases a steep diagonal line of face or even straight line of face has been selected.

Principles of Pillar extraction techniques

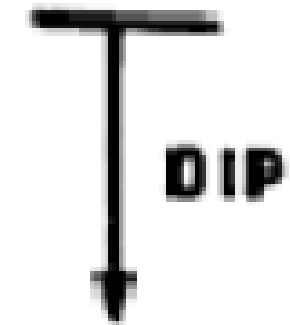
- ▶ Diagonal or step diagonal line of face provides protection as the working places are supported by solid pillars and also when the roof caves, there is less risk of goaf flushing into the working faces.
- ▶ The single lift extraction is limited to height of 4.8 m or less. If the thickness of the seam is more than 4.8 m, the extraction is done in multi-lifts and in that case hydraulic sand stowing is insisted upon. Seams up to 4.8 m thick can be mined by caving in one pass.
- ▶ Whatever the method of extraction, the working area is systematically supported by cogs and props.

Diagonal line of extraction

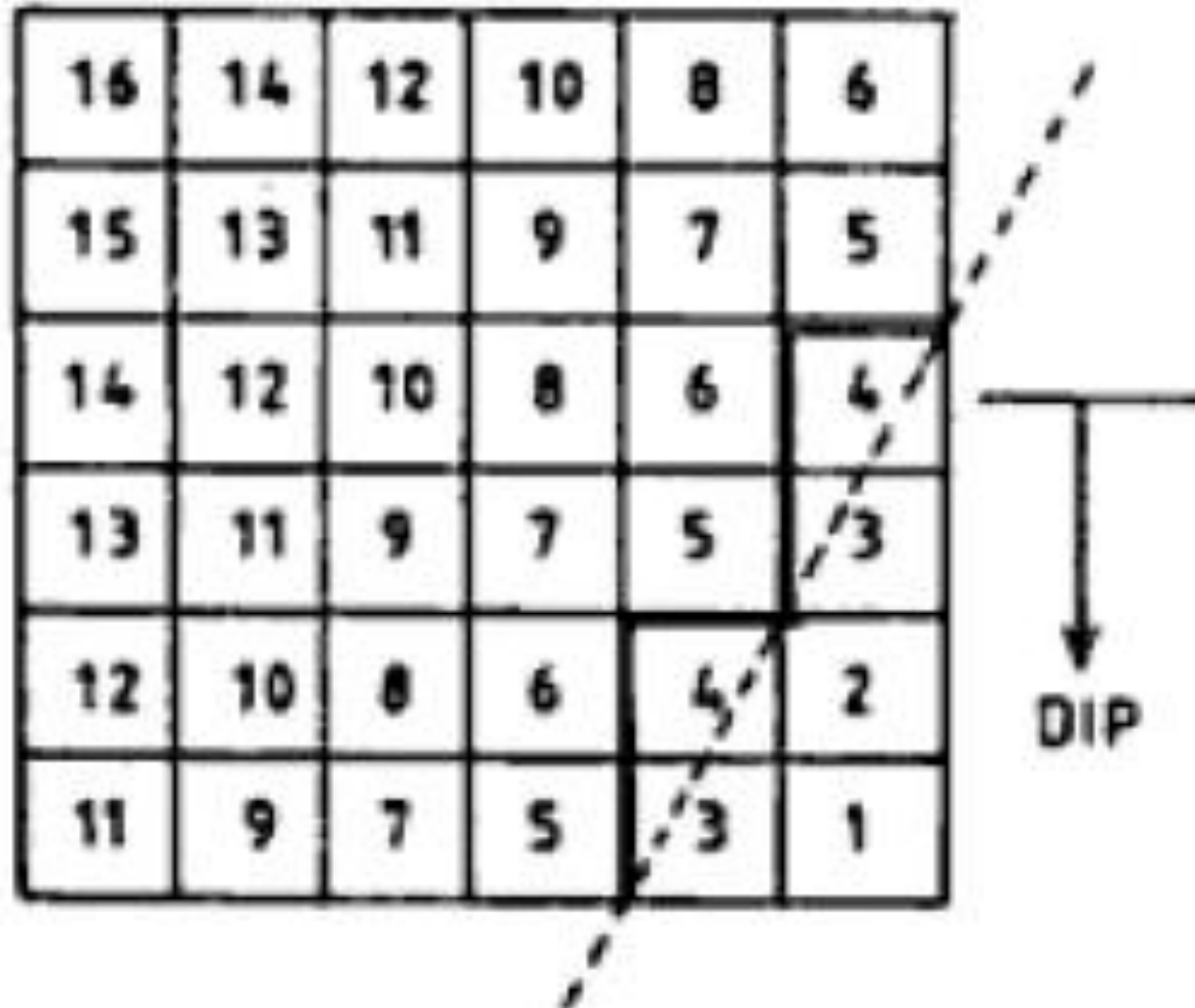


Step Diagonal line of extraction

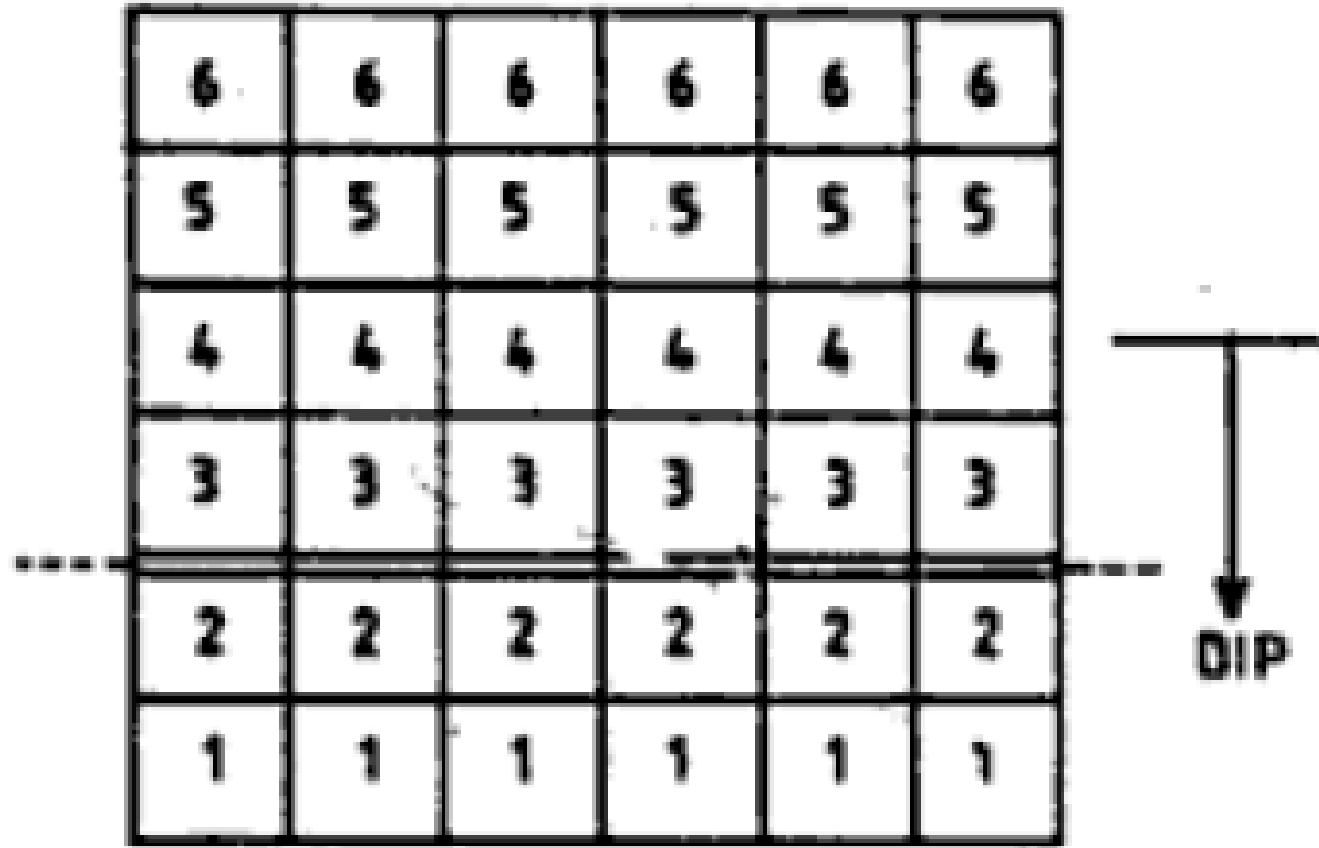
16	15	14	13	12	11
14	13	12	11	10	9
12	11	10	9	8	7
10	9	8	7	6	5
8	7	6	5	4	3
6	5	4	3	2	1



Steep diagonal line of extraction

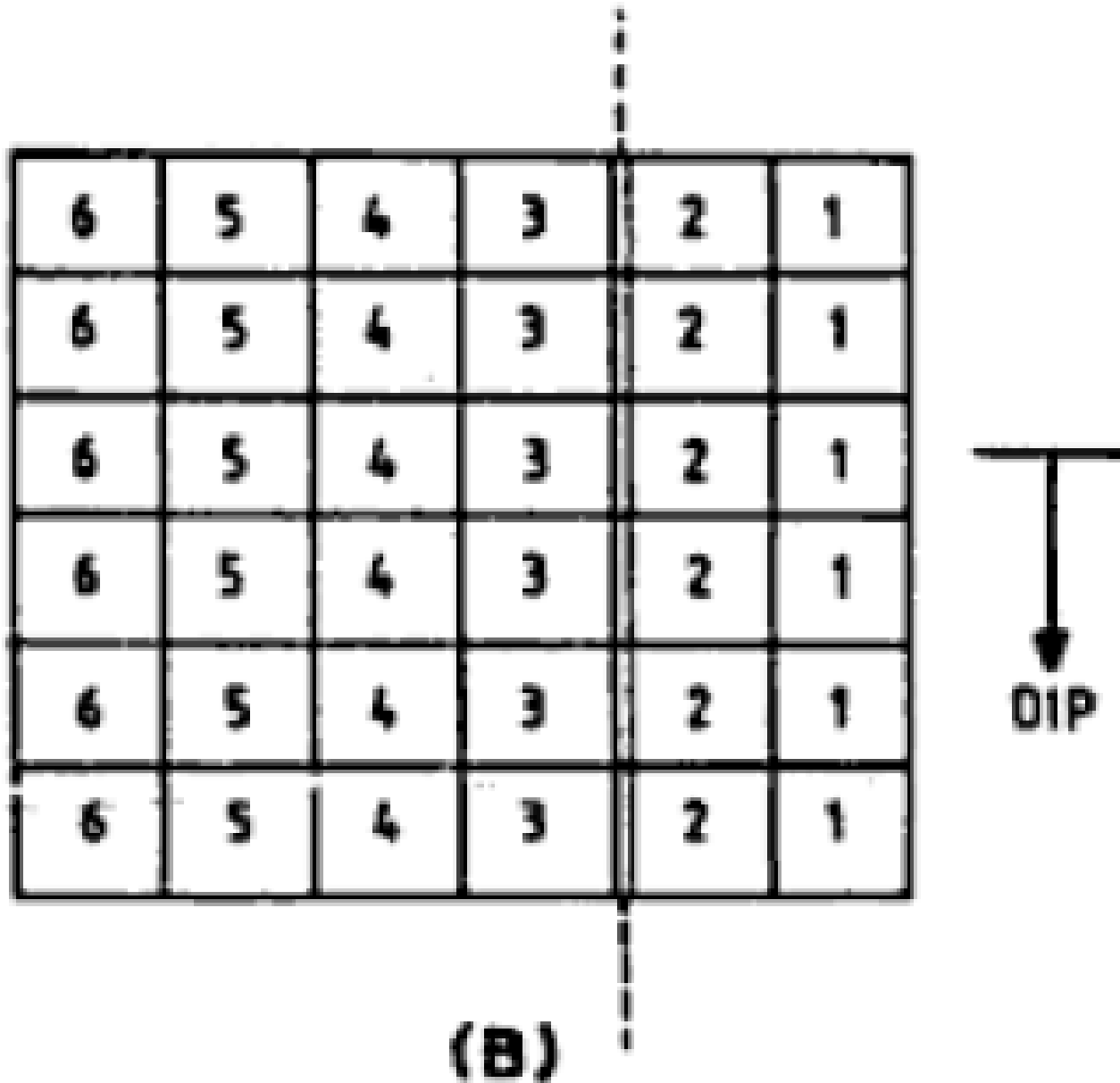


Straight line of extraction

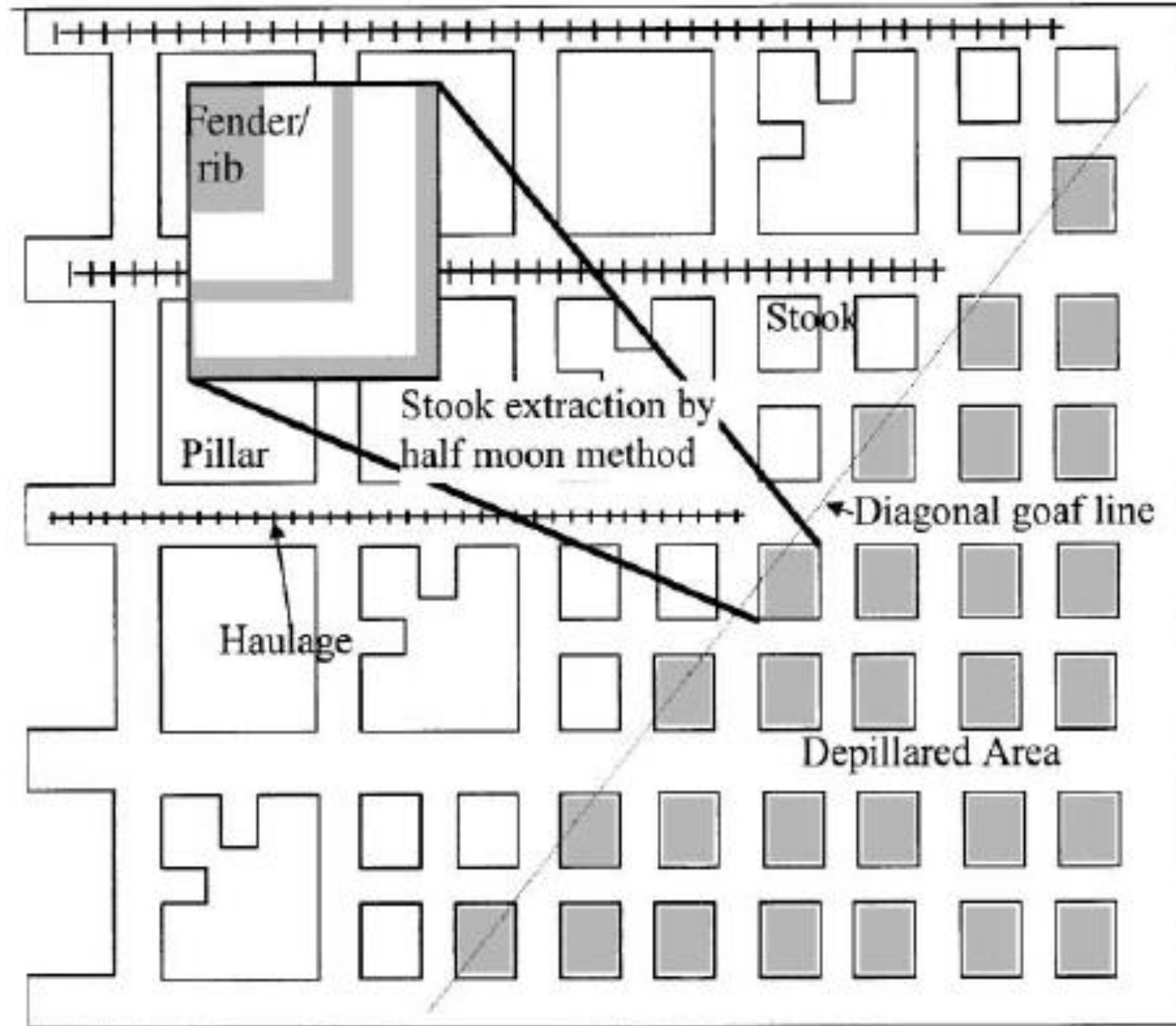


(A)

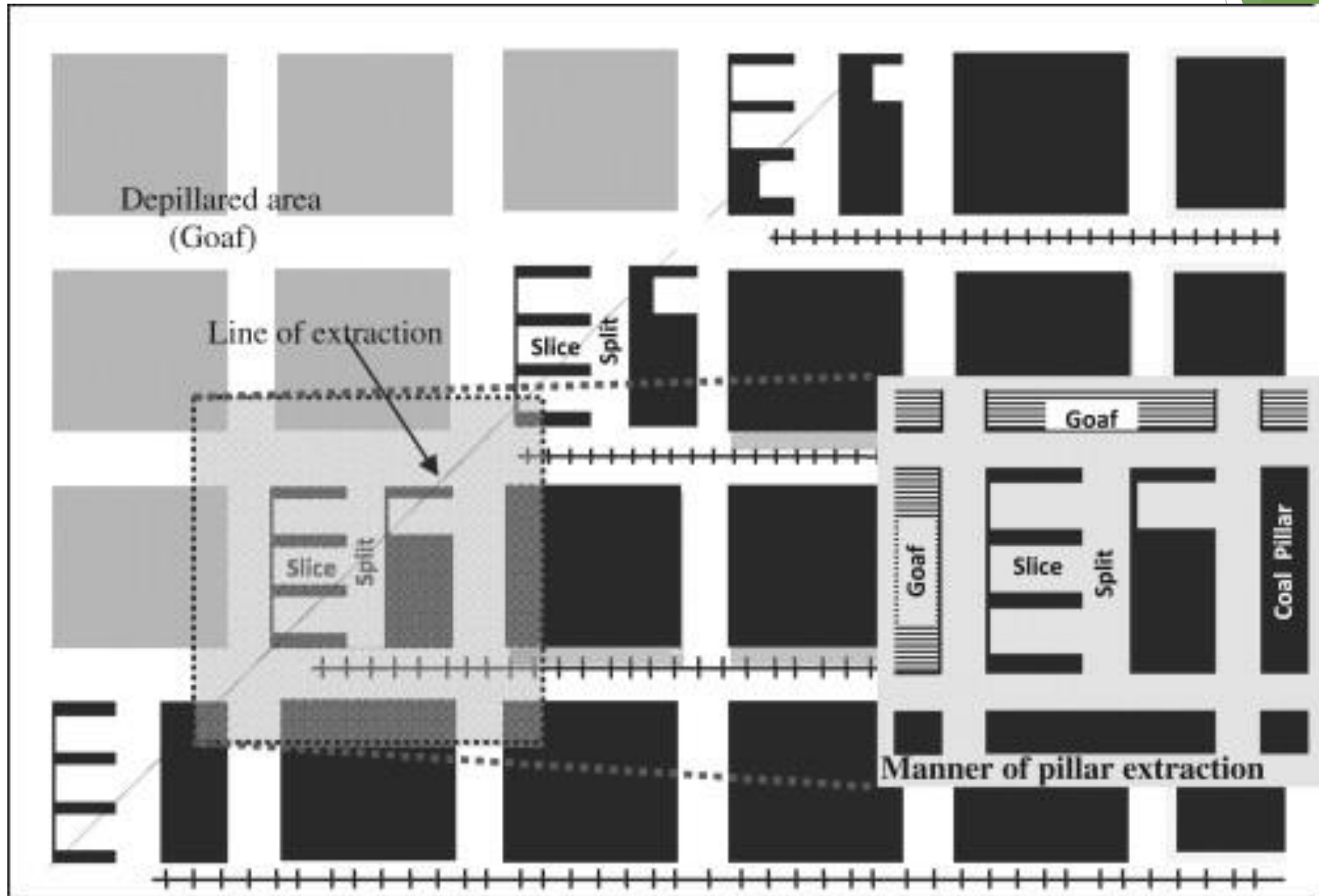
Straight line of extraction

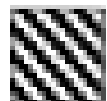
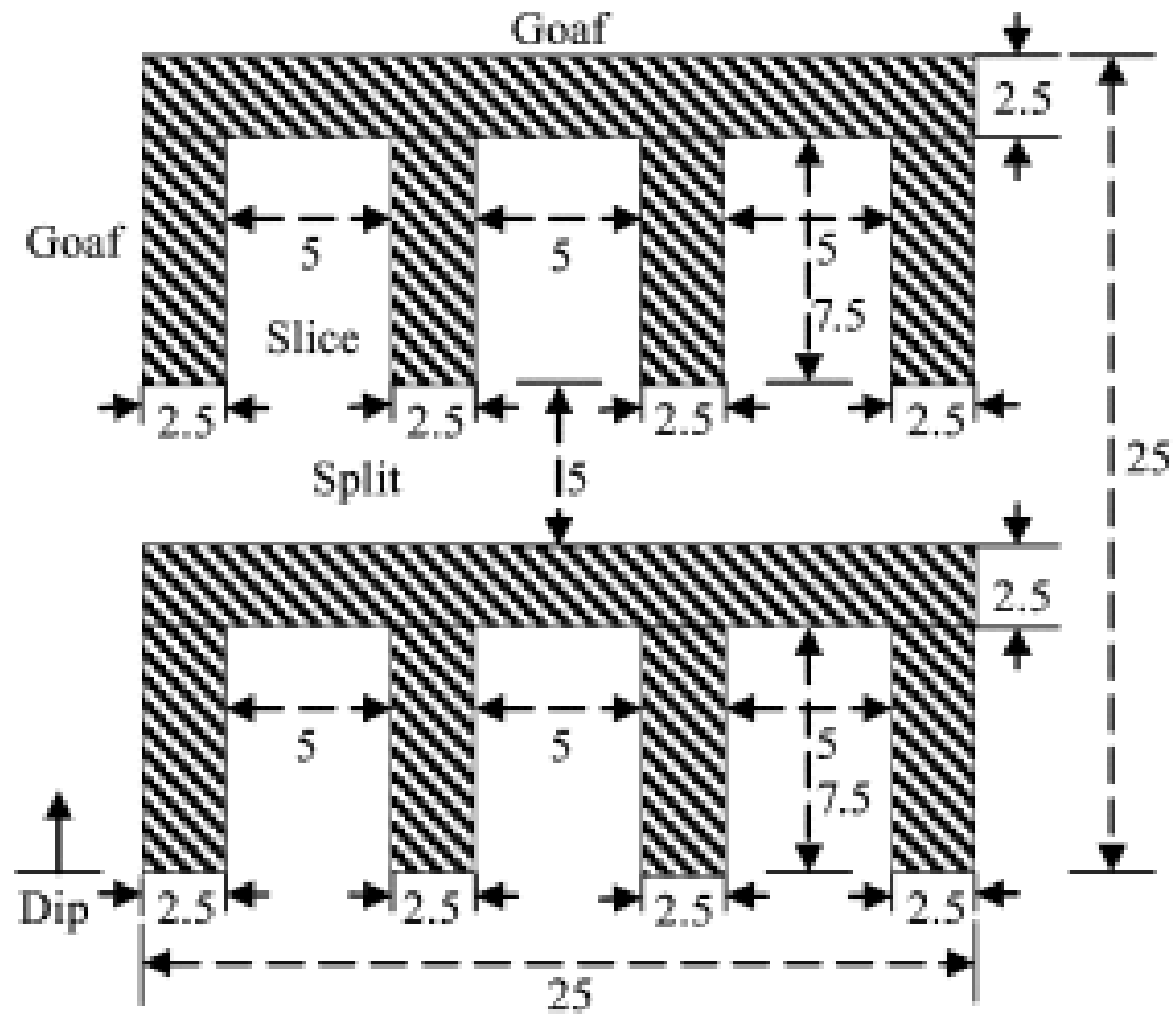


Stooking and slicing



Splitting and slicing





Rib/fender formed after splitting (width-5 m) and subsequent slicing (width-5 m)